## Investigation: BAHEMA SA (CVM code 701), fiscal year 2012, as it was on 2014-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Emp. Adm. Part. - Sem Setor Principal (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

## Document used

2012 DFP, version 1, filed on 2013-04-01; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2013-04-01).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=25807

## Possible manipulation (engine v1, study 1)

## Financial risk (context only)

Altman Z'' (financial risk, context only)
  Value: 10.636 (index)
  Components: X1 = 0.894; X2 = 0.285; X3 = −0.110; X4 = 4.361 (indices)
  Percentile in sector and year: 81 (n = 72)
  The company's own history: fewer than 5 previous fiscal years with a value
  - No cut-off zone: the available source does not define zones for Z''
  - Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
  Value: F = 4/9 (7 computable)
  Signals:
    - Return on assets (ROA): 0
    - Operating cash flow on assets (CFO): 1
    - Change in ROA (ΔROA): 1
    - Cash above earnings (ACCRUAL): 1
    - Change in leverage (ΔLEVER): 0
    - Change in current ratio (ΔLIQUID): 1
    - No equity issuance (EQ_OFFER): —
    - Change in gross margin (ΔMARGIN): —
    - Change in asset turnover (ΔTURN): 0
  - No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement
  - No long-term debt in either year: the change in leverage (ΔLEVER) is zero and the signal is 0 (project choice; the article only defines fell = 1, rose = 0, p. 8)

## Changes

Highlighted in the following year's comparative: none.
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets

of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 9 accounts above the R$1,000 tolerance (group a, totals and key lines: 0; group b, reclassifications: 9), in the annex at the end.

## What was known on the date

- DFP 2010, v1: 2011-03-04
- DFP 2011, v1: 2012-03-16
- FCA 2012, v1: 2012-03-16
- FCA 2012, v2: 2012-11-01
- DFP 2012, v1: 2013-04-01
- DFP 2013, v1: 2014-03-31

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis consolidadas acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira consolidada da BAHEMA S/A em 31 de dezembro de 2012, o desempenho consolidado de suas operações e os seus fluxos de caixa para o exercício findo naquela data, de acordo com as normas internacionais de relatório financeiro (IFRS) emitidas pelo International Accounting Standards Board - IASB e as práticas contábeis adotadas no Brasil."
Auditor in the 2011 FCA: HLB Audilink & Cia Auditores.
Auditor in the 2012 FCA: HLB Audilink & Cia Auditores.
Auditor in the 2013 FCA: HLB Audilink & Cia Auditores.
Auditor changes: none in the FCAs filed by the date.
Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project

and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-11T00:41:26+00:00.

## Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: BIOMA EDUCAÇÃO S.A..

## Annex: full list of differences in the following year's comparative (R$ million)

(b) DRE 3.04.02.12 Outras Despesas: 0.0 in the version used, −0.1 in the comparative (difference of −0.1)
(b) DRE 3.04.02.01 Gerais e Administrativas: −1.3 in the version used, −1.2 in the comparative (difference of 0.1)
(b) DRE 3.04.02.04 Remuneração do Pessoal: −0.6 in the version used, −0.6 in the comparative (difference of 0.1)
(b) DRE 3.04 Despesas/Receitas Operacionais: −6.9 in the version used, −6.9 in the comparative (difference of 0.0)
(b) DRE 3.04.02 Despesas Gerais e Administrativas: −3.6 in the version used, −3.6 in the comparative (difference of 0.0)
(b) DRE 3.05 Resultado Antes do Resultado Financeiro e dos Tributos: −6.9 in the version used, −6.9 in the comparative (difference of 0.0)
(b) DRE 3.06 Resultado Financeiro: 5.3 in the version used, 5.3 in the comparative (difference of −0.0)
(b) DRE 3.06.02 Despesas Financeiras: −0.7 in the version used, −0.7 in the comparative (difference of −0.0)
(b) DRE 3.06.02.02 Outras Despesas Financeiras: −0.7 in the version used, −0.7 in the comparative (difference of −0.0)